

NONG KHAI, Thailand

WMO: 483520

Lat: 17.8650N

Lon: 102.7472E

Elev: 576

StdP: 14.39

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	57.1	60.0	46.0	46.9	66.8	49.3	53.2	66.9	11.2	71.9	9.7	74.7	1.2	90	0.296

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.3	100.4	78.0	98.1	78.0	96.0	78.1	82.0	91.5	81.5	90.7	80.9	89.7	3.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.9	158.8	85.8	79.2	155.0	85.1	78.9	153.3	84.7	46.5	92.2	45.8	91.0	45.2	90.2	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.0	7.2	52.2	103.6	3.2	2.2	49.9	105.2	48.0	106.5	46.1	107.7	43.8	109.3
			49.1	83.7	3.1	1.3	46.9	84.7	45.1	85.4	43.4	86.1	41.1	87.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.1	73.9	78.0	82.8	86.2	85.2	84.3	83.3	82.7	82.8	81.9	78.5	74.1	
	DBStd	5.54	4.60	5.46	5.09	4.42	3.57	2.81	2.81	2.47	2.43	2.88	3.90	4.86	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	9	3	2	1	0	0	0	0	0	0	0	0	3	
	CDD50	11367	742	783	1015	1086	1091	1030	1033	1013	983	989	856	746	
	CDD65	5900	280	364	551	636	626	580	568	548	533	524	406	284	
	CDH74	64105	2231	3618	6441	8218	7532	6802	6261	5829	5732	5492	3697	2252	
	CDH80	27617	730	1635	3313	4529	3613	2934	2405	2076	2120	2152	1407	703	
Wind	WSAvg	1.9	2.0	2.1	2.1	2.0	1.9	1.7	1.6	1.7	1.7	1.9	2.0	2.1	
Precipitation	PrecAvg	67.4	0.4	0.7	1.8	3.5	9.2	10.9	12.5	13.4	11.1	3.7	0.7	0.2	
	PrecMax	88.1	2.9	3.1	5.1	9.7	18.2	20.0	21.1	25.9	20.0	9.5	3.5	1.6	
	PrecMin	47.9	0.0	0.0	0.0	1.0	2.6	3.8	7.2	7.9	5.1	0.1	0.0	0.0	
	PrecStd	9.6	0.7	0.7	1.5	2.2	3.5	3.3	3.6	4.0	3.7	2.4	1.1	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	97.1	101.3	104.4	101.8	97.9	96.4	94.1	93.8	93.9	94.0	91.4	
		MCWB	72.5	73.6	75.7	78.3	80.1	80.6	81.0	80.5	80.1	79.2	76.7	74.9	
	2%	DB	88.9	94.4	98.9	101.5	98.4	95.0	93.4	92.2	92.1	92.0	91.2	88.6	
		MCWB	71.9	73.3	75.6	78.5	79.8	80.6	80.6	80.3	79.5	77.9	75.9	73.5	
	5%	DB	86.7	92.2	96.5	98.9	95.8	92.9	91.4	90.5	90.6	90.5	89.3	86.4	
		MCWB	71.1	73.3	76.0	78.6	79.8	80.4	80.3	79.9	79.1	77.1	74.6	72.0	
	10%	DB	84.4	89.5	93.9	96.2	93.2	91.0	89.5	88.7	89.0	89.1	87.3	84.2	
		MCWB	70.0	73.0	75.7	78.3	79.4	80.1	79.9	79.6	78.8	76.5	73.5	70.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	78.0	80.5	81.8	82.4	82.8	82.7	82.4	81.9	81.0	79.0	77.4
			MCDB	86.4	90.9	94.6	96.1	94.9	92.2	91.8	90.1	90.1	90.5	89.3	87.2
2%		WB	74.4	76.5	78.7	80.8	81.4	81.9	81.6	81.4	81.0	79.9	77.7	75.3	
		MCDB	84.3	89.2	92.2	95.1	92.8	90.9	89.8	88.7	88.6	88.2	86.9	84.7	
5%		WB	72.8	75.3	77.6	79.9	80.7	81.2	80.9	80.7	80.4	79.1	76.6	73.7	
		MCDB	82.8	87.5	91.0	93.7	91.6	89.7	88.5	87.8	87.5	86.6	84.9	83.4	
10%		WB	71.4	74.1	76.7	79.1	80.1	80.7	80.3	80.1	79.8	78.3	75.5	72.4	
		MCDB	81.4	85.4	89.7	92.2	90.1	88.7	87.3	86.5	86.3	85.3	83.6	81.8	
Mean Daily Temperature Range	5% DB	MDBR	20.2	20.8	20.0	18.3	14.8	12.6	11.4	11.1	11.7	13.8	16.9	18.9	
		MCDBR	21.8	23.5	23.4	21.8	18.6	15.2	14.4	14.0	14.1	15.1	17.7	19.2	
		MCWBR	7.3	6.3	6.0	5.4	5.1	4.6	4.5	4.6	4.3	4.2	5.2	6.4	
	5% WB	MCDBR	19.3	20.0	20.4	19.5	16.3	13.8	12.9	12.7	12.7	13.4	15.6	17.7	
		MCWBR	7.3	6.3	6.2	6.0	5.3	5.0	4.7	4.9	4.7	4.3	5.3	6.3	
Clear-Sky Solar Irradiance	taub	0.466	0.575	0.691	0.654	0.504	0.458	0.446	0.464	0.507	0.542	0.455	0.460		
	taud	2.113	1.839	1.619	1.740	2.169	2.326	2.356	2.286	2.151	2.010	2.230	2.148		
	Ebn at Noon	258	235	213	223	256	265	269	266	255	240	259	255		
	Edh at Noon	48	66	84	75	49	41	40	43	49	55	42	45		
All-Sky Solar Radiation	RadAvg	1473	1608	1631	1776	1670	1531	1446	1426	1468	1542	1534	1429		
	RadStd	101	90	92	88	119	118	100	100	83	98	59	59		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page